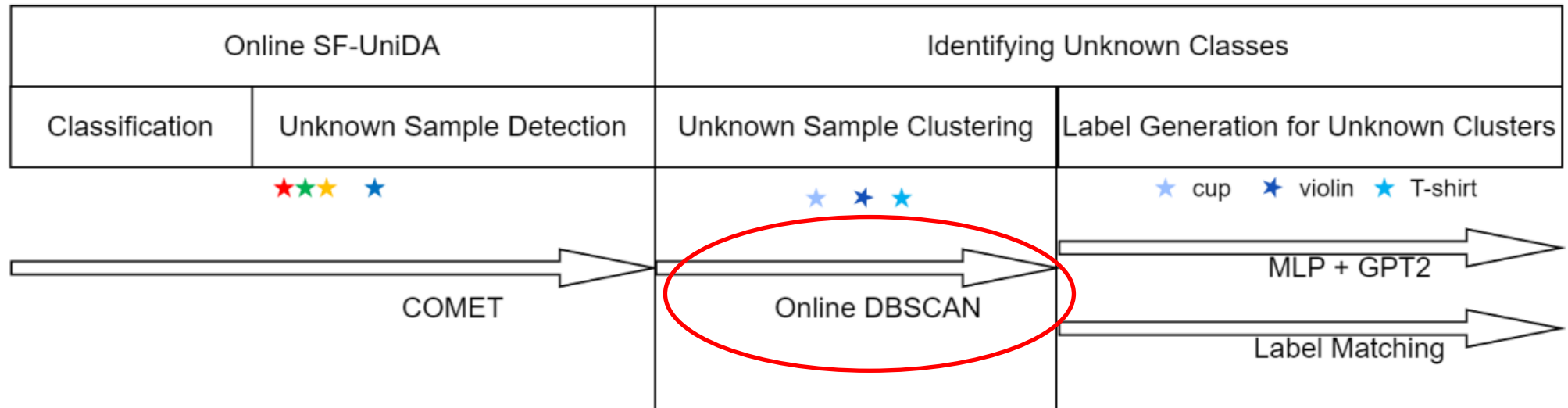
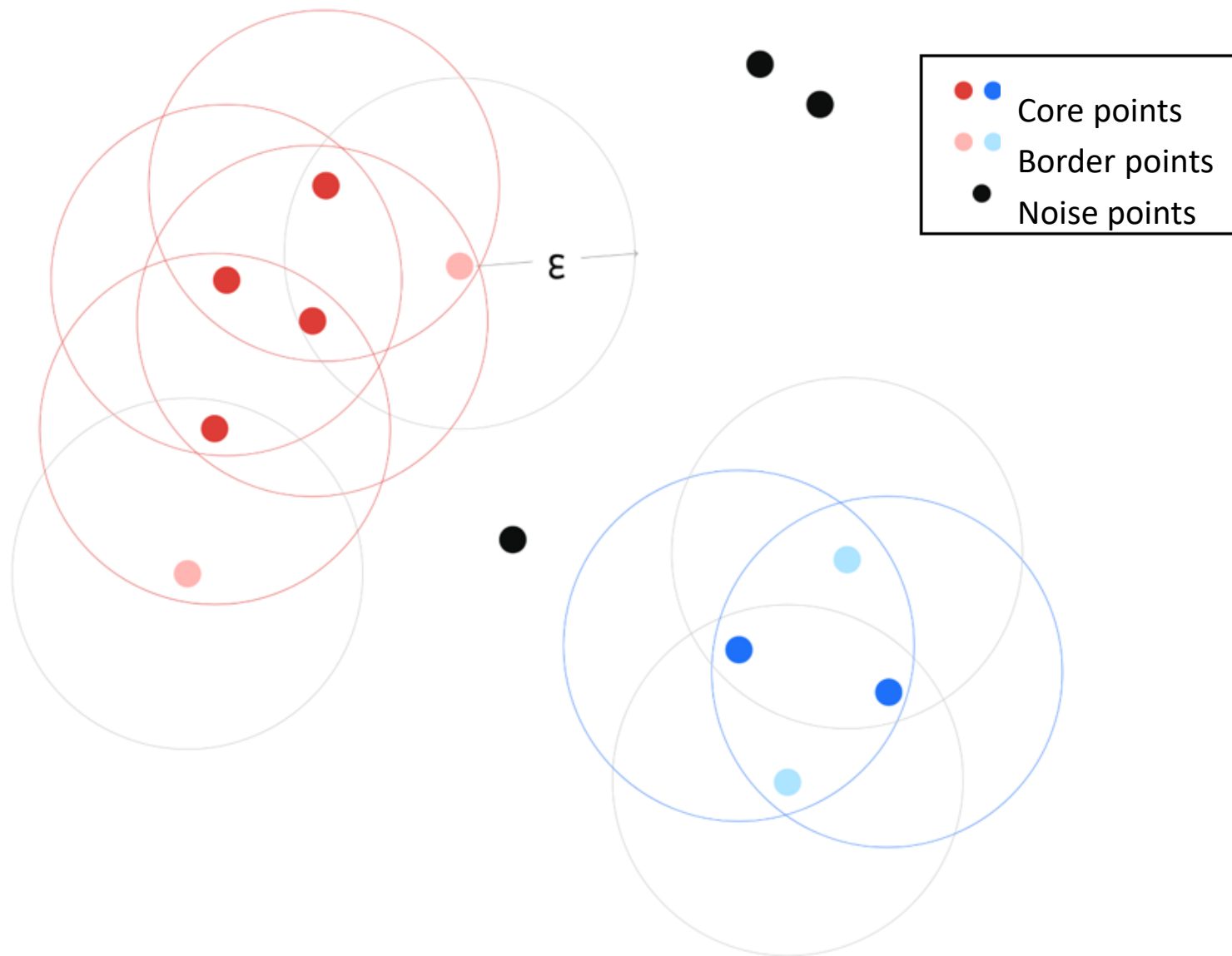

Identifying Unknown Classes in Universal Domain Adaptation

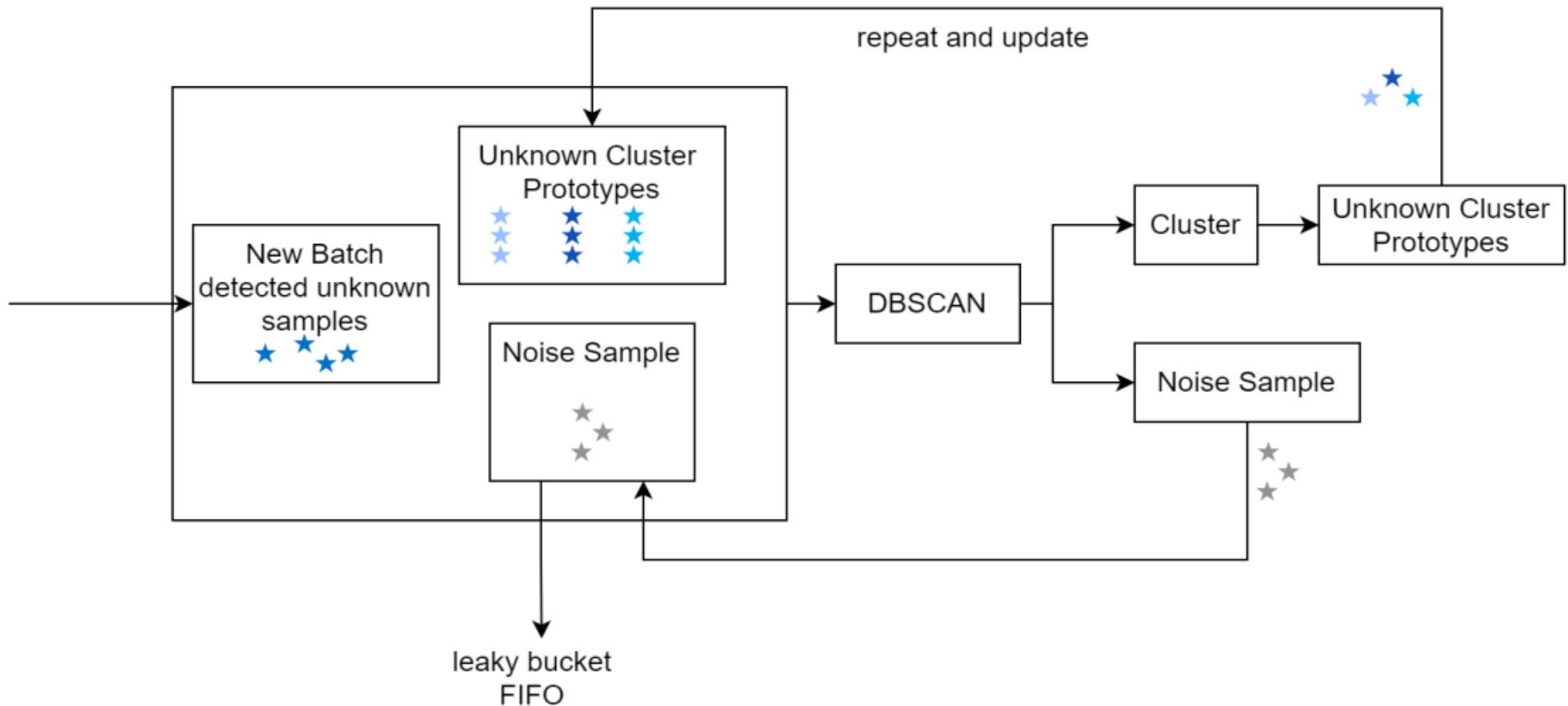
Research Thesis – Zhengyu Bao

Framework – Identifying Unknown Classes

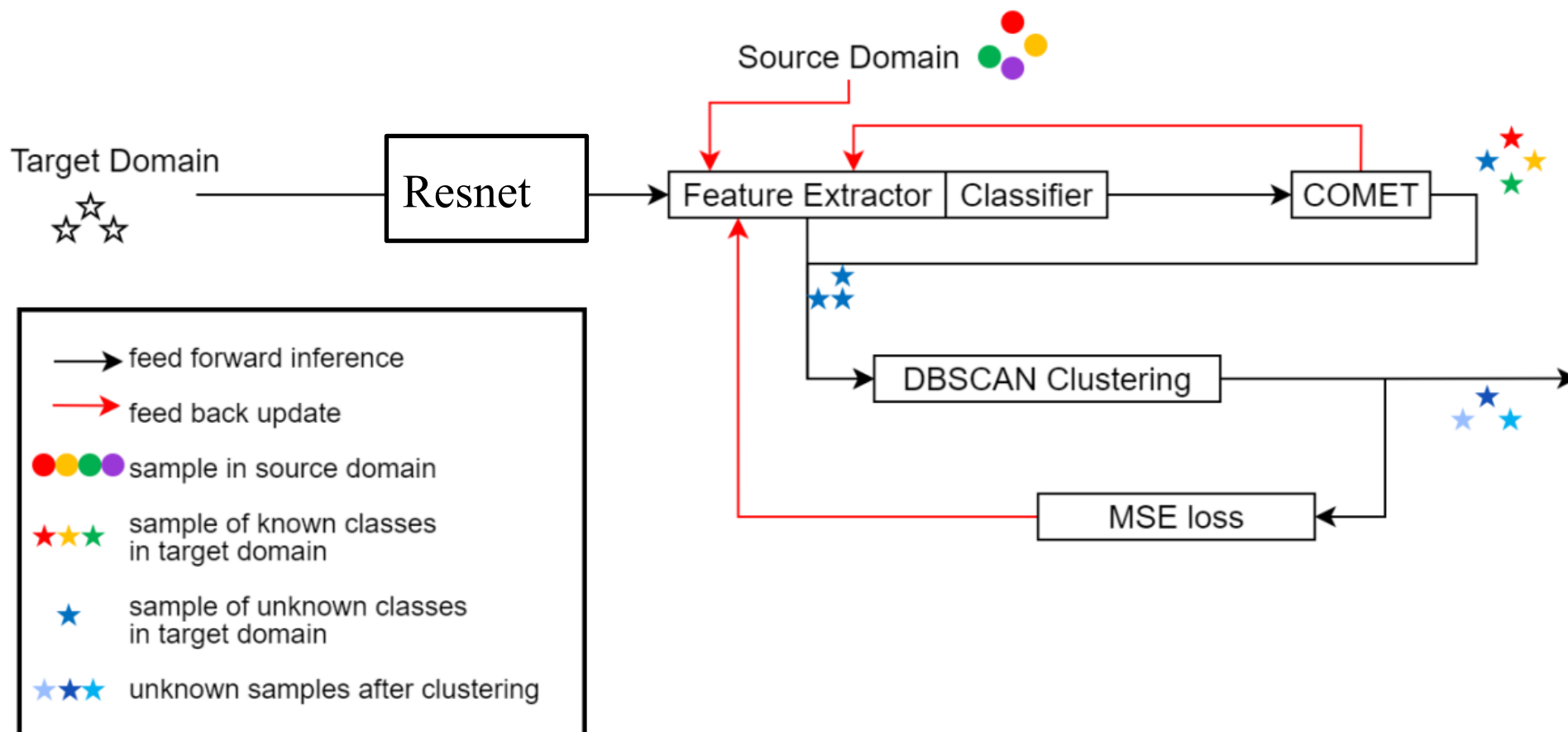




DBSCAN for Online Clustering

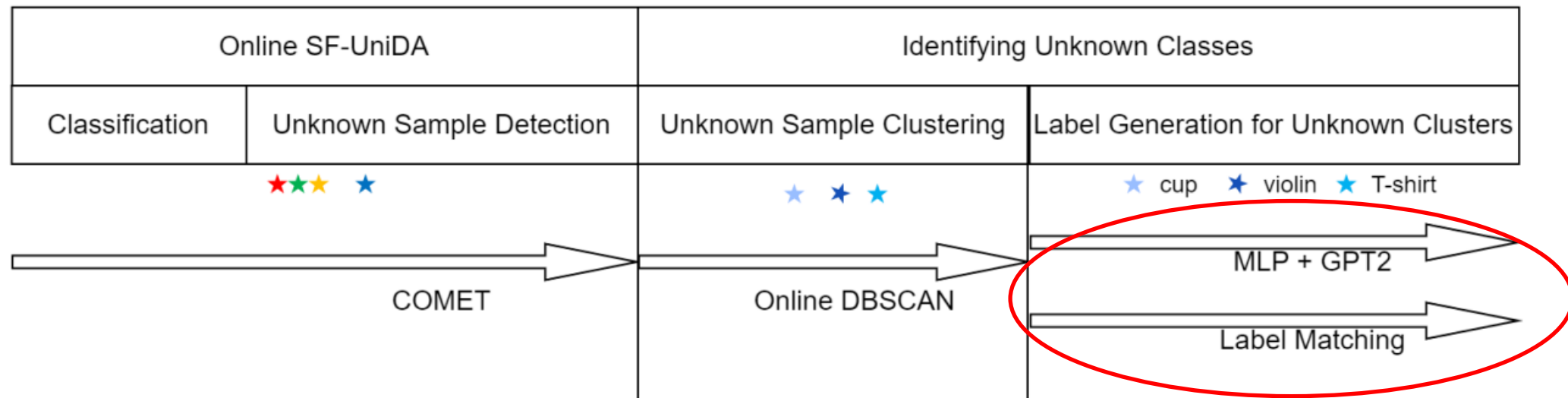


Clustering Result for Adaptation



Hyperparameters			UniDA acc			Unknown Clusters			
2. learning	ϵ	$n_{\min}$	known	unknown	H-score	precision	recall	f1-score	noise ratio
no	5	4	56.1%	67.3%	61.2%	86.4%	93.5%	89.8%	76.5%
yes	5	4	55.7%	67.3%	61.0%	86.4%	94.3%	90.2%	76.6%
no	6	4	55.9%	67.0%	60.9%	89.3%	63.3%	74.1%	71.6%
yes	6	4	55.7%	67.5%	61.0%	89.2%	61.8%	73.0%	71.6%

Framework – Identifying Unknown Classes



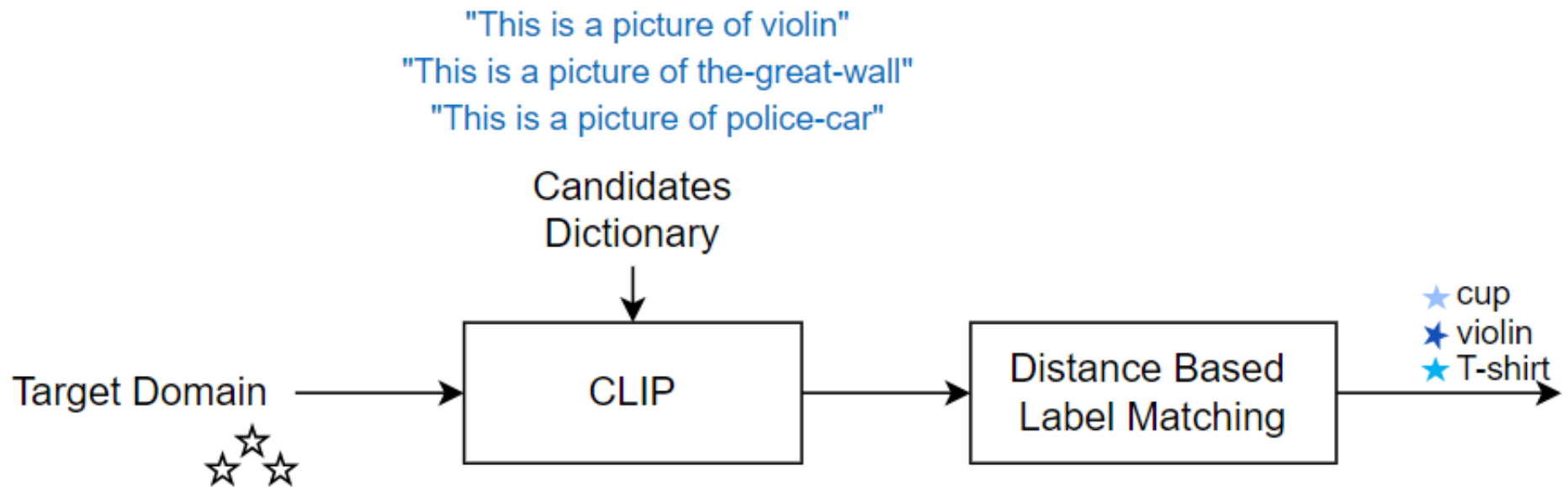


Table 4.1.: Labels cover rate of dataset by candidates

n-candidates	5000	10000	20000	40000
DomainNet	57.39%	69.28%	79.71%	86.09%

Table 4.2.: Top-3 accuracy of label matching

n-candidates	5000	10000	20000	40000
DomainNet-Real	40.25%	42.94%	37.86%	27.19%
DomainNet-Painting	29.25%	28.94%	25.78%	19.68%
Visda-train	37.21%	28.29%	11.33%	7.31%

$$P(\text{top-3 correct} \mid \text{true label covered by candidates}) = 40.25\% / 57.39\% = 70.13\%$$

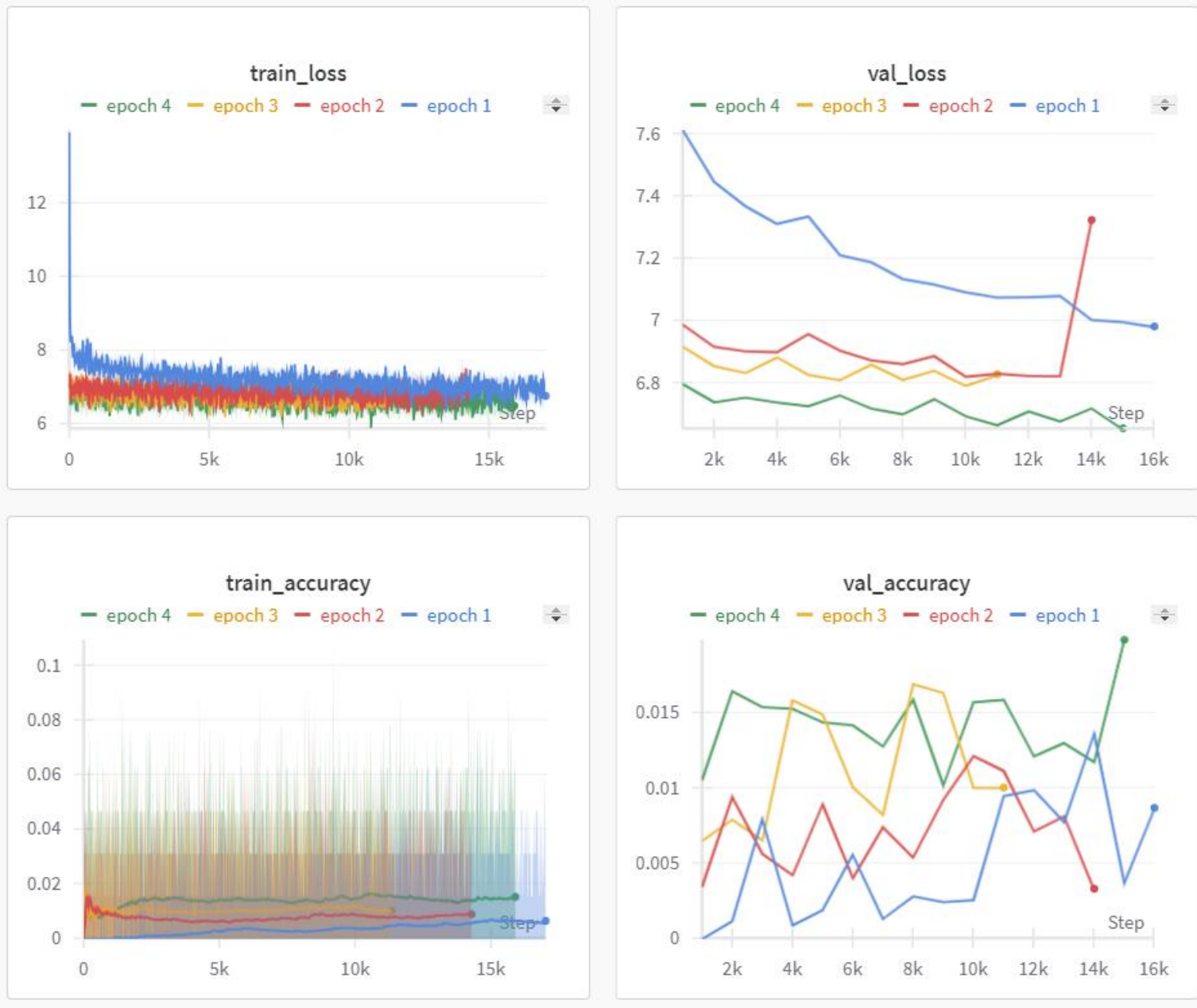
$$\mathbf{x}_1 = CLIP(\mathbf{x}_0) \quad \text{with : } \mathbf{x}_1 \in \mathbb{R}^{512}$$
$$\mathbf{x}_2 = MLP(\mathbf{x}_1) \quad \text{with : } \mathbf{x}_1 \in \mathbb{R}^{768 \times 10}$$

$$\mathbf{z}_1 = GPT2(\text{tokenize}(\text{'Please generate a label for the image:'}); \mathbf{x}_2) \quad \text{with : } \mathbf{z}_1 \in \mathbb{R}^{50257}$$
$$\mathbf{z}_2 = GPT2(\text{tokenize}(\text{'Please generate a label for the image:'}); \mathbf{x}_2; \mathbf{z}_1) \quad \text{with : } \mathbf{z}_2 \in \mathbb{R}^{50257}$$

CE Loss + Cosine Similarity Loss

Training based on ImageNet

Label Generation



Comparison – Label Generation vs. Selection



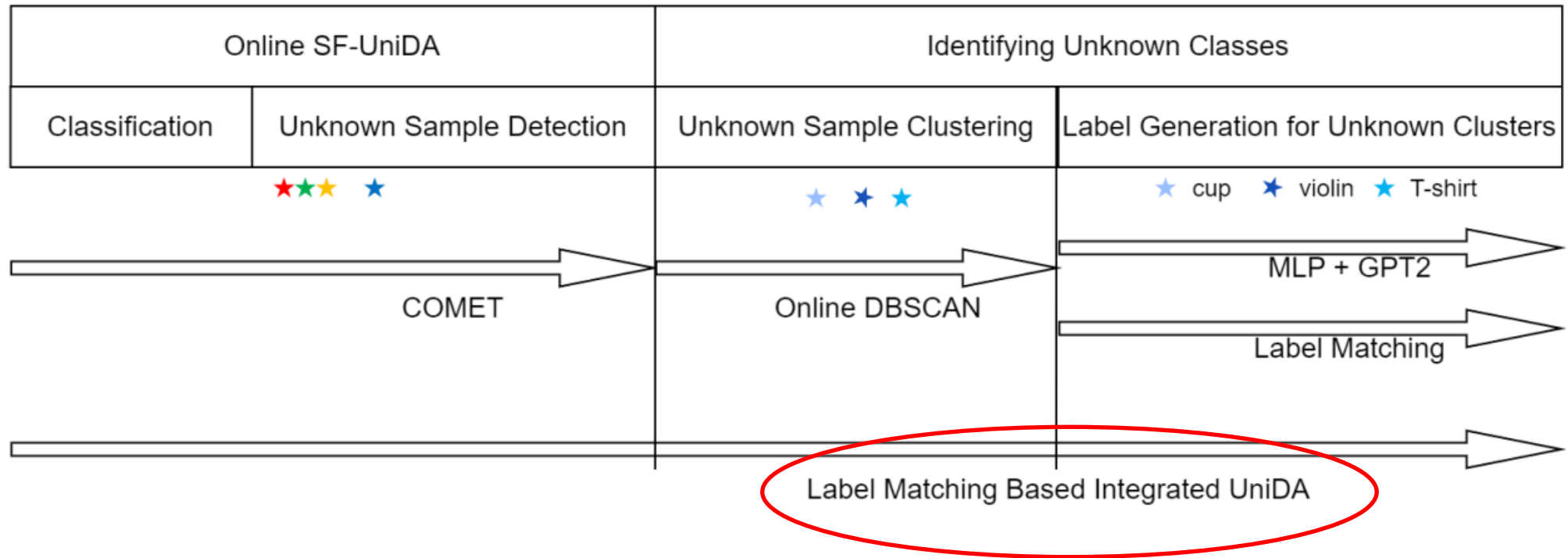
Label Generation vs. Selection

The only difference is scale!

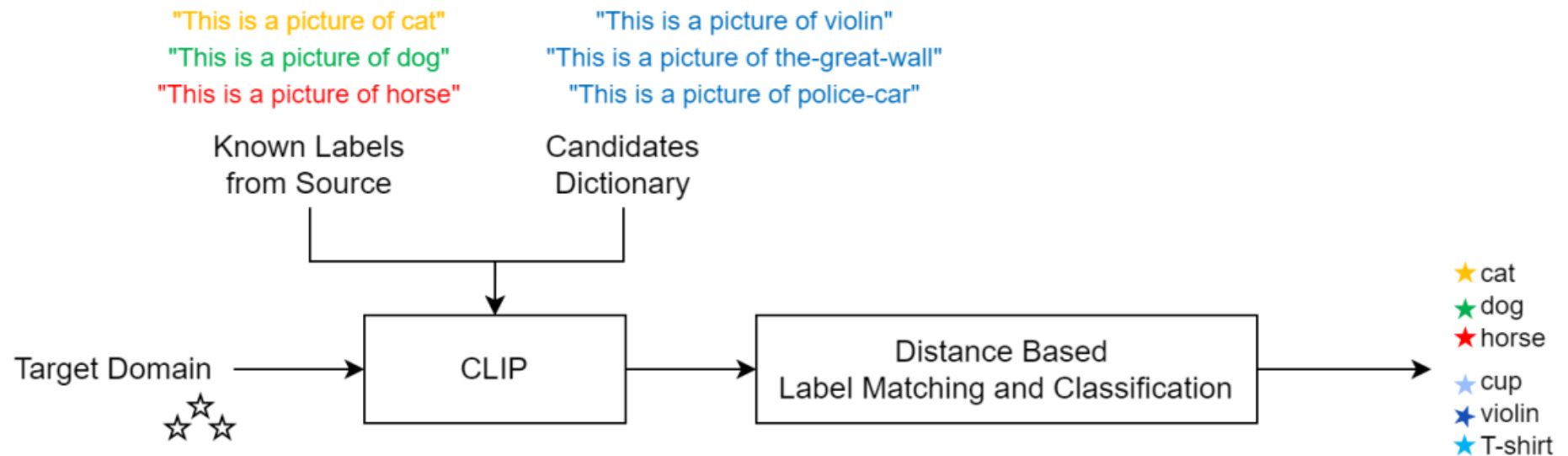
	GPT2 Generation	Label Selection
# Target Labels	50257*50257	20000
Architecture	LLM (Decoder-only Transformer)	Distance based
Training Effort	MLP 8M	No training
Training procedure	ImageNet	-

*Reference: Unconstrained open vocabulary image classification: Zero-shot transfer from text to image via clip inversion

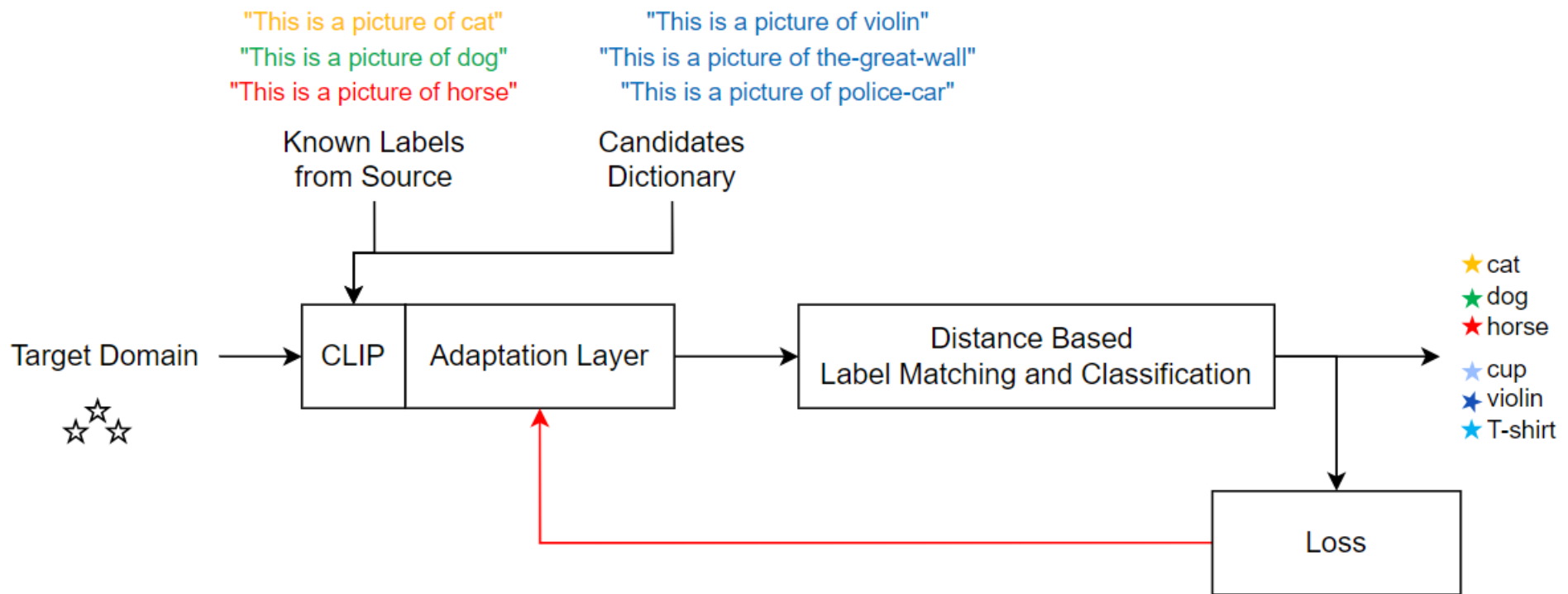
Framework – Identifying Unknown Classes



Integrated UniDA with Labal Matching



Integrated UniDA with Labal Matching



Linear Adaptation Layer

$$\mathbf{x}_2^t = \mathbf{W}\mathbf{x}_1^t + \mathbf{b}$$

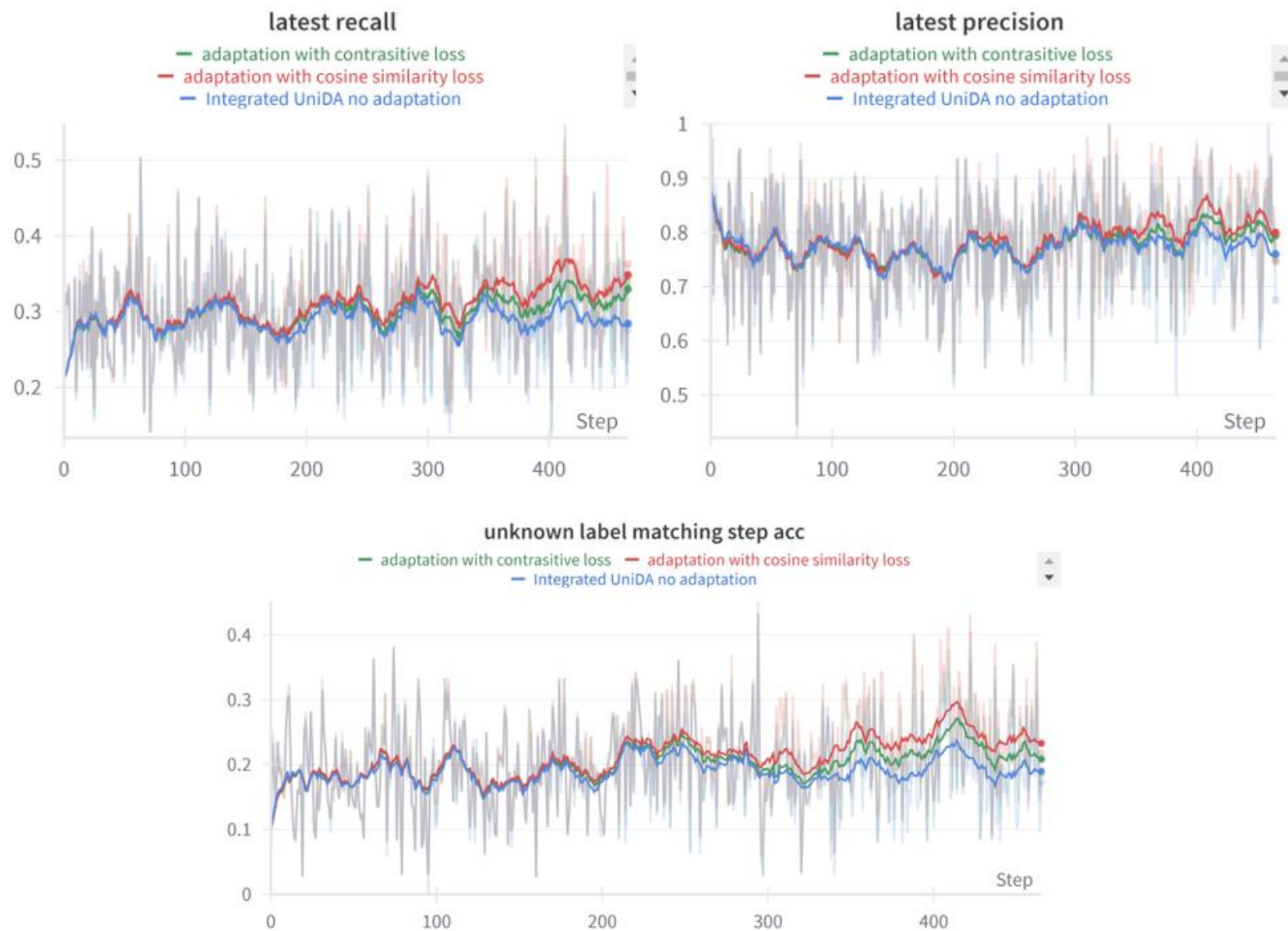
with : $\mathbf{x}_1^t, \mathbf{x}_2^t, \mathbf{b} \in \mathbb{R}^{512}$ and $\mathbf{W} \in \mathbb{R}^{512 \times 512}$

Special Initialization Method

$$\mathbf{W} = \mathbf{I} \in \mathbb{R}^{512 \times 512}$$

$$\mathbf{b} = \mathbf{0} \in \mathbb{R}^{512}$$

Effect of Adaptation Layer



Comparison – one-step vs. two-step



Table 6.1.: one-step vs. two-step

method	known	UniDA acc		Unknown Clusters				
		unknown	H-score	precision	recall	f1-score	noise ratio	label acc
DomainNet painting to real								
one-step1	68.0%	84.7%	75.4%	78.5%	31.0%	44.4%	0%	21.3%
two-step1	55.7%	67.3%	61.0%	86.4%	94.3%	90.2%	76.6%	-
two-step2	55.7%	67.5%	61.0%	89.2%	61.8%	73.0%	71.6%	-
DomainNet real to painting								
one-step1	36.8%	90.7%	52.4%	26.7%	11.9%	16.5%	0%	11.6%
two-step1	38.4%	86.1%	53.1%	21.5%	67.9%	32.7%	84.3%	-
VisDA								
one-step2	77.7%	78.8%	78.2%	90.6%	4.8%	9.1%	0%	1.6%
one-step3	88.3%	55.5%	68.2%	96.7%	45.4%	61.8%	0%	38.2%
two-step3	57.0%	64.4%	60.5%	55.3%	86.0%	67.3%	54.2%	-
two-step4	56.9%	64.8%	60.6%	27.0%	82.2%	40.7%	31.1%	-

Comparison – Adaptation Efficiency



	Dense layer	Linear Adapter
# Parameter	+	+
Adaptation Efficiency	-	+
Adaptation Capability	non-linear	linear
Zero-shot Learning Ability	-	+

$$\mathbf{x}_2 = \text{ReLU}(W_1 \mathbf{x}_1 + b_1) - \text{ReLU}(W_2 \mathbf{x}_1 + b_2) + \mathbf{x}_1$$

with initialization: $W_1 = W_2, b_1 = b_2$

Conclusion

